A2.2 Cell structure

NAME	DATE	TOTAL
		_____ / 48

About 50 minutes, or one section at a time. A calculator is allowed. The number of marks shows how many separate points to make.

Section A: Recall

9 MARKS

1 **LIST** three structures found in all cells. [A2.2.4] [3]

2 **STATE** the size of ribosomes in prokaryotes, and in eukaryotes. [A2.2.5, A2.2.6] [2]

3 **STATE** the function of the Golgi apparatus. [A2.2.6] [1]

4 **STATE** the main component of cell walls in plants, and in fungi. [A2.2.8] [2]

5 **STATE** one cell type with many nuclei. [A2.2.9] [1]

Section B: Understanding and drawing

17 MARKS

6 **EXPLAIN** why electron microscopes show more detail than light microscopes. [A2.2.3] [2]

7 **OUTLINE** how immunofluorescence shows the location of one type of protein in a cell. [A2.2.3] [2]

8 **DRAW** a prokaryotic cell as seen in an electron micrograph, and label five structures. [A2.2.5] [5]

9 **EXPLAIN** how Paramecium carries out four functions of life. [A2.2.7] [4]

10 **COMPARE AND CONTRAST** the structure of a typical plant cell with that of a typical animal cell. [A2.2.8] [4]

Section C: Working with data

14 MARKS

All values in this section are invented for practice.

11 [A2.2.2]

a A sperm tail is 50 µm long, and a student draws it 75 mm long. **CALCULATE** the magnification. [2]

b A micrograph at ×400 shows a cell 28 mm wide. **CALCULATE** its actual width in µm. [2]

c A scale bar labelled 10 µm measures 25 mm. **CALCULATE** the magnification. [2]

d On the same image, a mitochondrion is 6 mm long. **CALCULATE** its actual length. [2]

e The graticule in an eyepiece has 100 divisions. It lines up with 1 mm on a stage micrometer. A cell spans 30 divisions. **CALCULATE** the width of the cell. [2]

12 [A2.2.10]

An unknown cell in an electron micrograph is 2 µm long. No membrane surrounds its DNA, and it has no mitochondria. It has a thick outer layer.

a **DEDUCE** the type of cell, giving two reasons. [2]

b **NAME** the region in which its DNA is found, and the type of its ribosomes. [2]

Section D: Longer answer

8 MARKS

13 **COMPARE AND CONTRAST** prokaryotic and eukaryotic cells. [A2.2.4–6] [8]